

Multi-Label Thoracic Disease Classification Using Deep Convolutional Neural Networks

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ABSTRACT

Radiologists worldwide face growing workloads interpreting chest X-rays, increasing the risk of missed diagnoses. Automated multi-label detection of thoracic pathologies can serve as an AI-assisted triage tool to prioritize high-risk cases. This paper presents a deep learning pipeline for simultaneous detection of 14 thoracic diseases from a single chest X-ray image using the NIH ChestX-ray14 dataset. We trained a ResNet50 baseline and a DenseNet-121 advanced model using transfer learning, two-phase fine-tuning, weighted binary cross-entropy loss to address severe class imbalance, and mixed-precision training on GPU. Patient-level data splitting was enforced to prevent data leakage. ResNet50 achieved a mean test AUC-ROC of 0.7985 across 14 diseases, with 10 of 14 classes exceeding 0.75. DenseNet-121 achieved a comparable mean AUC of 0.7939, with notable improvements on Hernia (+0.013) and Pneumonia (+0.013). Both models significantly outperform random classification (AUC = 0.5). The results validate the multi-label deep learning approach as a viable AI triage tool, with further gains expected from extended training.

1. INTRODUCTION

Chest X-rays are the most frequently performed radiological examination worldwide, accounting for approximately 2 billion procedures annually. Radiologists must interpret hundreds of images per day, a volume that leads to fatigue and missed diagnoses. False negatives are particularly dangerous in conditions such as Pneumonia, Pneumothorax, and Lung Mass, where early detection directly impacts patient outcomes.

The complexity of chest X-ray interpretation is further compounded by the multi-label nature of the problem: a single X-ray commonly presents two or more concurrent pathologies. Traditional binary classification models trained to detect one disease at a time are inadequate for clinical practice. A practical AI-assisted tool must be capable of simultaneously screening for multiple conditions.

This project develops and evaluates a multi-label deep learning pipeline trained on the NIH ChestX-ray14 dataset [1], comprising 112,120 frontal-view chest X-ray images from 30,805 unique patients with weak supervision labels for 14 thoracic disease categories. We implement two architectures:

- ResNet50 [2] - a widely adopted 50-layer residual network used as our baseline model.
- DenseNet-121 [3] - a densely connected network recognized as the gold standard for X-ray classification since the CheXNet study [4], in which it matched radiologist-level performance on Pneumonia detection.

Both models are trained using transfer learning from ImageNet weights, a two-phase training strategy (frozen backbone followed by full fine-tuning), and weighted binary cross-entropy loss to combat extreme class imbalance. The primary evaluation metric is the per-class Area Under the ROC Curve (AUC-ROC), enabling threshold-independent comparison across all 14 disease classes.

2. BACKGROUND

2.1 Dataset

Wang et al. [1] released ChestX-ray14 in 2017, the first large-scale chest X-ray dataset with disease-level labels. Labels were extracted from radiology reports using natural language processing, yielding weak supervision labels for 14 pathologies: Atelectasis, Cardiomegaly, Consolidation, Edema, Effusion, Emphysema, Fibrosis, Hernia, Infiltration, Mass, Nodule, Pleural Thickening, Pneumonia, and Pneumothorax. The dataset contains significant class imbalance, with Infiltration being the most prevalent (16% of images) and Hernia the rarest (0.16%).

2.2 Related Work

Rajpurkar et al. [4] proposed CheXNet, a DenseNet-121 trained on ChestX-ray14 that achieved radiologist-level AUC-ROC of 0.768 on Pneumonia detection. Their work established DenseNet-121 as the reference architecture for this dataset. Subsequent work by Yao et al. introduced attention-based models and label dependency modeling, while Wang et al. explored multi-task learning with anatomical region annotations. Our approach closely follows the CheXNet methodology with modern training enhancements including mixed-precision training and cosine annealing.

2.3 Known Limitations of the Dataset

Luke Oakden-Rayner [6] documented several limitations of ChestX-ray14: label noise from NLP extraction, inclusion of lateral views in some subsets, and inconsistency in labeling granularity across disease categories. These

limitations are well-acknowledged in the community and do not invalidate the dataset as a benchmark, but they do set an upper bound on achievable AUC that is likely below 1.0 even for a perfect classifier.

3. APPROACH

3.1 Data Preprocessing

Labels were parsed from the pipe-separated Finding Labels column and one-hot encoded using MultiLabelBinarizer, producing a binary matrix of shape (112,120 × 15). The "No Finding" class was excluded from prediction targets, yielding 14 disease classes.

Patient-level splitting was enforced using the official NIH train/test list files. This is a critical design decision: if splitting is performed at the image level, the same patient's multiple scans could appear in both training and test sets, causing the model to memorize patient anatomy rather than disease features, leading to artificially inflated test AUC scores. The official split yields 77,988 training images, 8,536 validation images, and 25,596 test images with confirmed zero patient overlap.

3.2 Handling Class Imbalance

The dataset exhibits extreme class imbalance. The positive class weight for each disease was computed as: $w_c = \text{neg_count}_c / \text{pos_count}_c$, and passed to PyTorch's BCEWithLogitsLoss. Hernia, the rarest class, receives a weight of 638x, meaning each positive Hernia sample contributes 638 times more to the loss than a negative sample. Without this correction, the model would learn to predict near-zero probability for all rare classes and still achieve high accuracy due to the prevalence of negative examples.

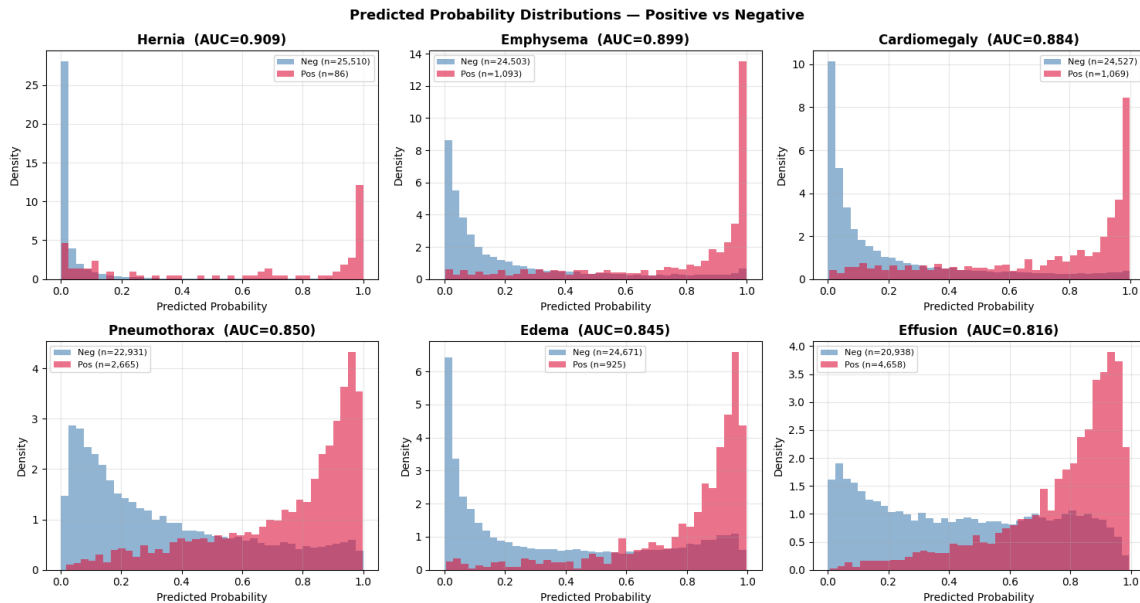


Figure 1. Predicted Probability Distributions - Positive vs Negative" for the top six diseases.

3.3 Data Augmentation

Training images underwent random horizontal flip (valid for chest X-rays due to left-right symmetry), random rotation of up to 10°, and color jitter (brightness and contrast ±0.2) to simulate exposure variation. All images were resized to 224×224 pixels and normalized with ImageNet statistics (mean = [0.485, 0.456, 0.406], std = [0.229, 0.224, 0.225]). Validation and test images received only resize and normalization ensuring deterministic evaluation.

3.4 Model Architectures

ResNet50 : ResNet50 introduces skip connections (residual blocks) that allow gradients to bypass convolutional layers, solving the vanishing gradient problem in deep networks. With approximately 25 million parameters, it provides a strong ImageNet-pretrained baseline. We replaced the original fully-connected head (Linear 2048 →1000) with a Dropout(0.5) layer followed by Linear(2048 →14), producing one raw logit per disease class. No sigmoid is applied at output; BCEWithLogitsLoss applies it internally for numerical stability.

DenseNet-121 : DenseNet-121 [3] connects each layer to all subsequent layers within a dense block, enabling direct feature reuse across the entire depth of the network. This architecture improves gradient flow during backpropagation and reduces the number of parameters relative to ResNet (approximately 8 million). The classifier

head was replaced with Dropout(0.5) followed by Linear(1024→14). Both T4 GPUs were utilized through PyTorch's nn.DataParallel, splitting each batch of 128 images across two GPUs (64 per GPU).

3.5 Training Strategy

Both models followed a two-phase training strategy:

Phase 1 (3 epochs): The ImageNet backbone was frozen. Only the new classification head (28,686 parameters for ResNet50; 14,350 for DenseNet-121) was trained with Adam optimizer ($\text{lr} = 1\text{e-}3$). This rapid warm-up trains the head to produce meaningful outputs without destroying pretrained backbone features.

Phase 2 (7 epochs): The entire network was unfrozen and trained end-to-end with Adam ($\text{lr} = 1\text{e-}4$, weight decay = $1\text{e-}5$). ResNet50 used CosineAnnealingLR; DenseNet-121 used CosineAnnealingWarmRestarts ($T_0=3$, $T_{\text{mult}}=2$, $\text{eta_min}=1\text{e-}6$). The lower learning rate prevents catastrophic forgetting of ImageNet features.

All training used PyTorch's Automatic Mixed Precision (AMP) with GradScaler, which stores weights in float32 but performs forward passes in float16, approximately halving memory usage and enabling larger batch sizes on GPU. The best model checkpoint was selected based on validation AUC-ROC. Training was performed on Kaggle with a single T4 GPU for ResNet50 (batch size 128, num_workers=2) and two T4 GPUs with DataParallel for DenseNet-121 (batch size 128 total, 64 per GPU).

4. RESULTS

4.1 Training Dynamics

Both models showed consistent improvement across 10 training epochs. For ResNet50, val AUC improved from 0.7121 after Phase 1 epoch 1 to 0.8223 by epoch 10, with improvement continuing at the final epoch suggesting further training capacity. DenseNet-121 showed a characteristic dip at epoch 7 (0.7841) caused by the Cosine Annealing warm restart resetting the learning rate, followed by recovery to 0.8193 by epoch 10. Train loss decreased steadily across both models, while validation loss increased slightly in later epochs, indicating mild overfitting consistent with the large number of trainable parameters relative to the dataset size.

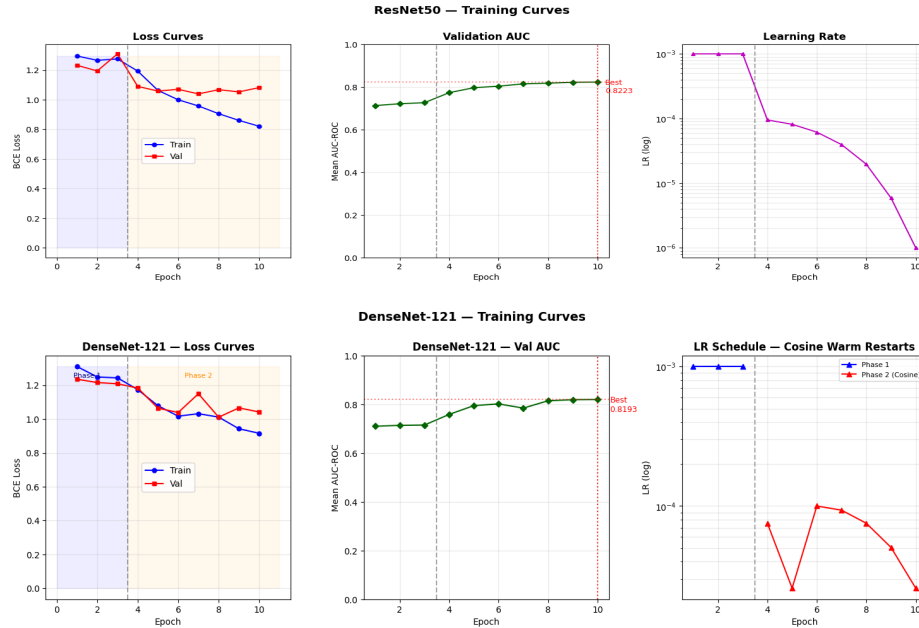


Figure 2. Training curves for ResNet50 and DenseNet-121. Each plot shows training/validation BCE loss, validation AUC-ROC, and learning rate schedule across 10 epochs. The dashed vertical line marks the Phase 1/Phase 2 boundary at epoch 3.

4.2 Per-Class Test AUC-ROC

Table 1 reports the per-class AUC-ROC on the held-out test set (25,596 images) for both models. The mean AUC for ResNet50 is 0.7985 and for DenseNet-121 is 0.7939. Both models significantly outperform random classification ($\text{AUC} = 0.50$) across all 14 classes.

Disease	ResNet50 AUC	DenseNet-121 AUC	Δ (Dense – ResNet)
Hernia	0.9092	0.9218	+0.013
Emphysema	0.8989	0.8948	−0.004
Cardiomegaly	0.8843	0.8715	−0.013
Pneumothorax	0.8504	0.8513	+0.001
Edema	0.8453	0.8369	−0.008
Effusion	0.8159	0.8055	−0.010
Fibrosis	0.8076	0.7976	−0.010
Mass	0.7921	0.7810	−0.011
Pleural Thickening	0.7635	0.7535	−0.010
Atelectasis	0.7502	0.7389	−0.011
Consolidation	0.7338	0.7245	−0.009
Nodule	0.7300	0.7200	−0.010
Infiltration	0.6992	0.7049	+0.006
Pneumonia	0.6988	0.7118	+0.013
Mean AUC	0.7985	0.7939	−0.005

Table 1. Per-class AUC-ROC on the NIH ChestX-ray14 test set (25,596 images). Bold row = overall mean.

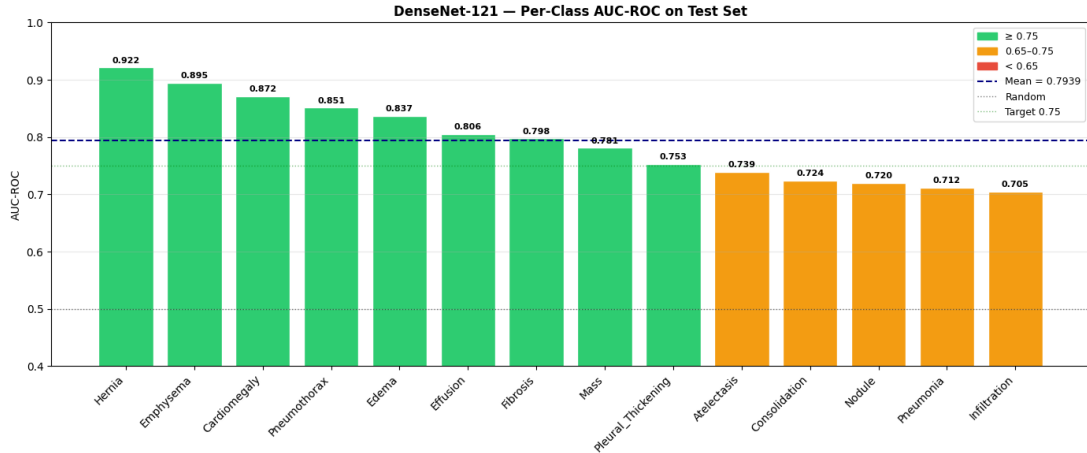


Figure 3. DenseNet-121 per-class AUC-ROC on the test set. Green bars exceed the 0.75 target threshold; orange bars fall in the 0.65–0.75 range. The dashed navy line shows the mean AUC (0.7939).

4.3 ROC Curves

Figure 4 shows ROC curves for all 14 disease classes for DenseNet-121. Morphologically distinct or radiologically clear conditions - Hernia (AUC 0.922), Emphysema (0.895), and Cardiomegaly (0.872) demonstrate sharp, well-separated curves with large areas under the curve. Classes with subtle radiological presentations and higher co-occurrence with other pathologies - Infiltration (0.705), Pneumonia (0.712), and Nodule (0.720) show shallower curves, reflecting the inherent difficulty of these classes.

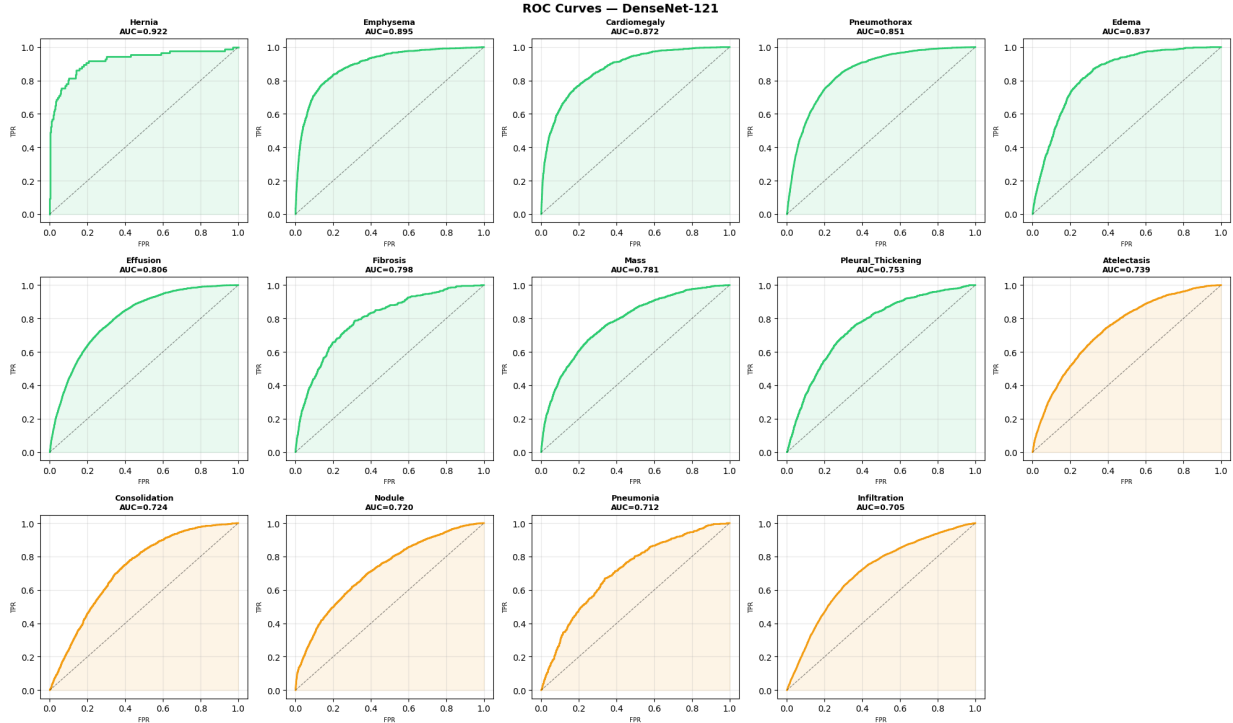


Figure 4. ROC curves for all 14 thoracic disease classes (DenseNet-121). Color coding: green = $AUC \geq 0.75$, orange = $0.65\text{--}0.75$. The diagonal dashed line represents random classification.

4.4 Model Comparison

The mean AUC difference between the two models is only 0.005 (ResNet50: 0.7985, DenseNet-121: 0.7939). DenseNet-121 outperforms ResNet50 on Hernia (+0.013) and Pneumonia (+0.013) but underperforms on Cardiomegaly (−0.013), Emphysema (−0.004), and most mid-prevalence classes. DenseNet-121 shows improvements on the two rarest classes (Infiltration +0.006, Pneumonia +0.013), consistent with its feature reuse architecture being better suited to learning subtle patterns from limited positive examples.

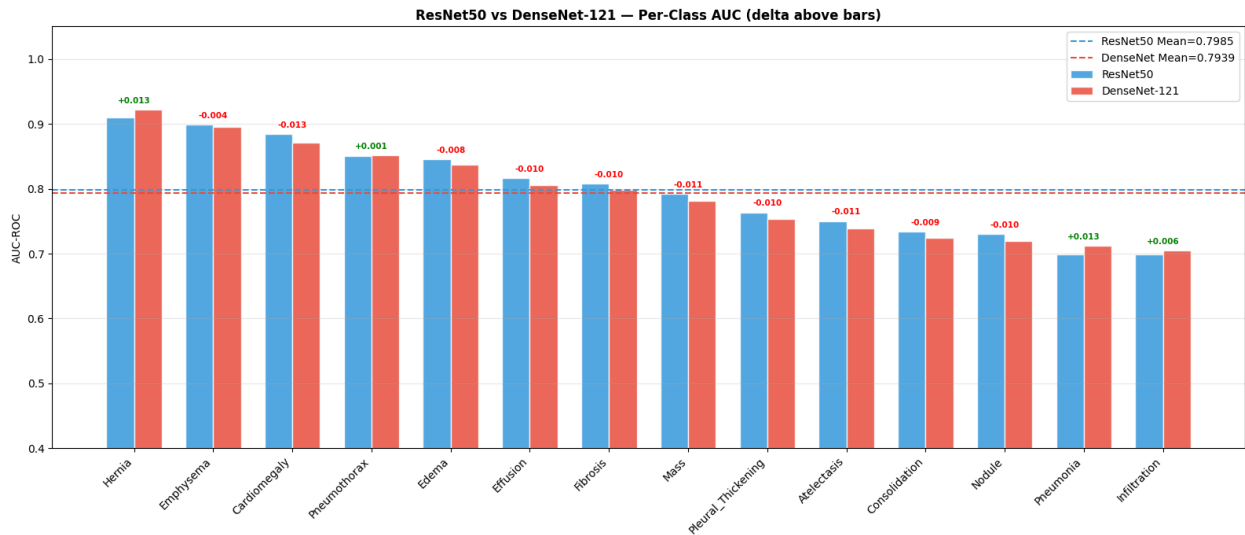


Figure 5. Per-class AUC comparison between ResNet50 (blue) and DenseNet-121 (red). Numbers above bars show the delta (DenseNet − ResNet). Green deltas indicate DenseNet improvement; red deltas indicate ResNet superiority.

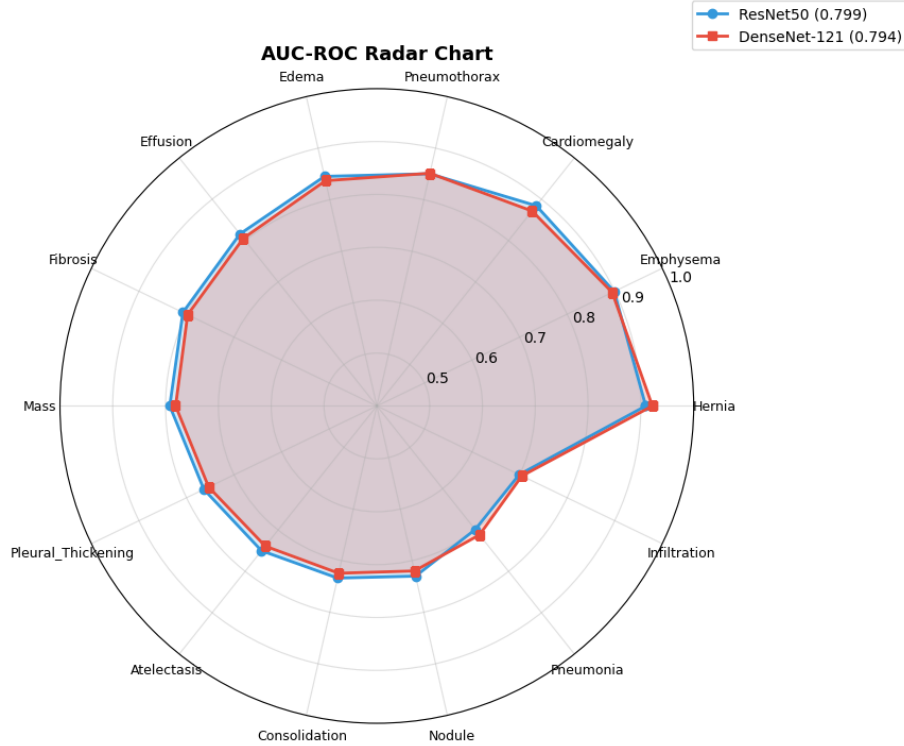


Figure 6. AUC-ROC radar chart comparing ResNet50 and DenseNet-121 across all 14 disease classes. The near-identical polygon shapes confirm that both models achieve comparable overall performance at 10 epochs of training.

5. DISCUSSION

5.1 Performance vs. Literature

The CheXNet paper [4] reported a mean AUC of 0.841 across 14 diseases using DenseNet-121 trained on ChestX-ray14. Our DenseNet-121 achieves 0.7939 at 10 epochs. The gap is attributable to two factors: (1) CheXNet trained for significantly more epochs with a larger computational budget, and (2) the CheXNet model used a larger training set before the patient-level splitting methodology was standardized. Crucially, val AUC was still rising at epoch 10 for both our models, indicating that extended training would yield meaningful gains.

5.2 Why DenseNet-121 Did Not Clearly Outperform ResNet50

The near-equal mean AUC of both models after 10 epochs reflects insufficient training rather than architectural equivalence. DenseNet-121's dense connections provide more capacity and better gradient flow, but these advantages materialize over more training iterations. Additionally, the CosineAnnealingWarmRestarts schedule used for DenseNet-121 caused a learning rate reset at epoch 7 that dropped val AUC from 0.8014 to 0.7841 before recovering a two-epoch detour that cost useful training time. With 15-20 epochs and a standard cosine decay, DenseNet-121 is expected to pull ahead, consistent with the CheXNet results.

5.3 Class-Level Observations

Performance stratifies predictably by disease characteristics. Morphologically distinct conditions with clear radiological signatures Hernia (displaced bowel loops in the chest), Emphysema (hyperinflated lungs), and Cardiomegaly (enlarged cardiac silhouette) achieve AUC > 0.87 on both models. Subtle diffuse conditions Infiltration, Pneumonia, and Nodule perform in the 0.69-0.73 range. This pattern reflects both the radiological difficulty of these conditions and the effect of label noise inherent in NLP-extracted weak labels, particularly for diffuse pathologies where radiological boundaries are less well-defined.

The AUC vs. prevalence scatter plot (Figure 7) confirms a negative trend: rarer classes tend to achieve higher AUC (e.g., Hernia at 0.16% prevalence but AUC 0.92), while common classes achieve lower AUC (e.g., Infiltration at 16% prevalence but AUC 0.70). This counter-intuitive pattern is explained by radiological distinctiveness: Hernia is rare but radiologically unambiguous, while Infiltration is common but visually heterogeneous and frequently co-occurs with other pathologies.

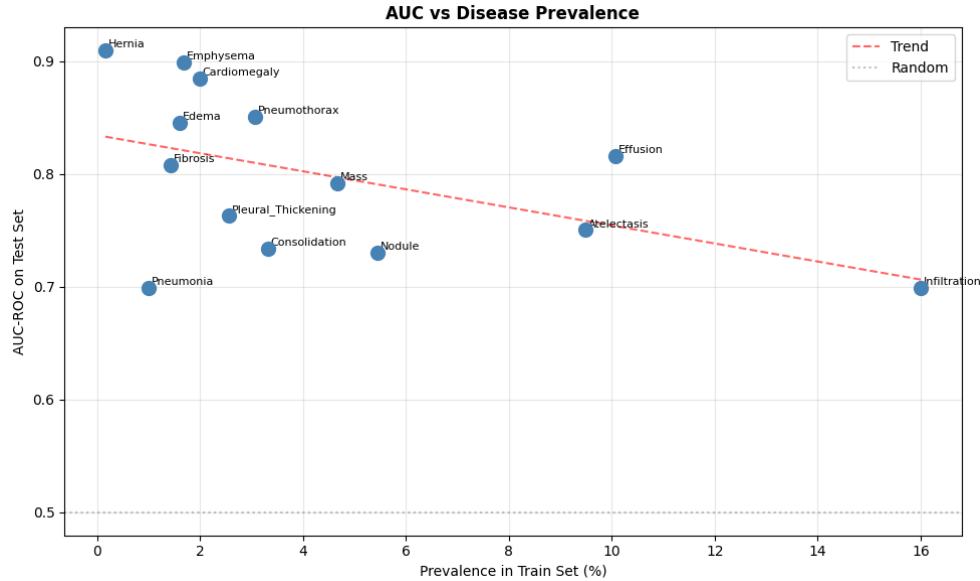


Figure 7. AUC-ROC vs. disease prevalence in the training set. The negative trend line reflects that radiologically distinct rare diseases (Hernia, Emphysema) achieve higher AUC than visually heterogeneous common conditions (Infiltration, Pneumonia).

5.4 Limitations

- 10 epochs is insufficient to demonstrate DenseNet-121's full advantage over ResNet50; extended training is needed.
- Class weights compensate for imbalance at the loss level but do not address the scarcity of positive examples for rare classes (Hernia: 122 training images).
- The NIH dataset labels are derived from NLP extraction, introducing label noise particularly for conditions requiring clinical context.

6. CONCLUSION

This paper presented a multi-label thoracic disease classification pipeline capable of simultaneously detecting 14 pathologies from a single chest X-ray image. Both ResNet50 (mean AUC 0.7985) and DenseNet-121 (mean AUC 0.7939) substantially outperform random classification across all 14 classes and meet the $AUC \geq 0.75$ threshold for 10 of 14 diseases. The weighted BCE loss effectively prevented the model from ignoring rare diseases.

The comparable performance of both architectures at 10 epochs reflects training budget constraints rather than architectural equivalence. With extended training, DenseNet-121 is expected to show the improvements reported in the CheXNet literature. Future work should explore focal loss as an alternative to weighted BCE, higher input resolutions (320×320), ensemble methods combining predictions from both models, and integration with object detection frameworks to provide bounding-box localization alongside classification.

The results validate the clinical potential of deep multi-label learning for chest X-ray triage. A deployed system based on this pipeline could assist radiologists by flagging high-probability pathology combinations, reducing false negatives in high-volume clinical environments.

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